

AquaSentinel

Robotics | Sanjivani Hero's Journey | AG26-1500\$41

Meet the Team



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Project Lead



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Problem & Opportunity

Problem Statement

A spill can pass in hours. Routine sampling may miss it.

55

polluted river stretches in Maharashtra, most in India

9.98 MLD

untreated sewage into the Godavari at Kopargaon

1× / month

CPCB river sampling, at best

Opportunity / Design Gap

THE GAP

No current option combines continuous river monitoring + low cost + proof.

✓ reads every 15 min

✓ ≈ ₹1.3 lakh a float*

✓ seals a sample on alert

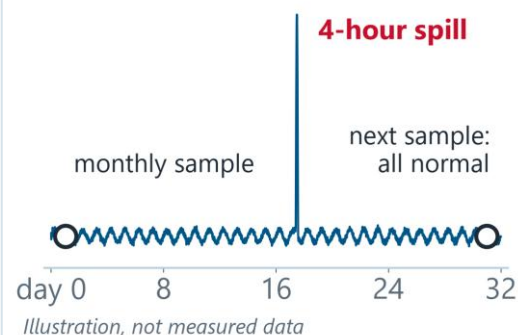
* prototype estimate

Existing Solutions

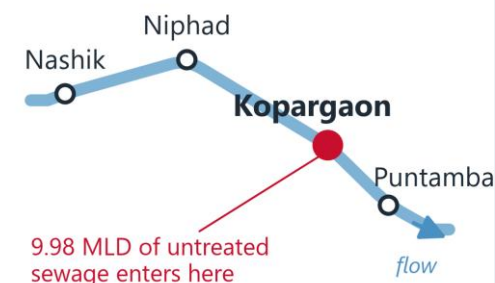
	24 × 7	Affordable	Keeps proof
Grab sample + lab results in days	✗	✓	✓
Real-time station ₹35 lakh+ each	✓	✗	✗
Drone boat fights the current	✗	—	—

Monthly checks miss a spill

conductivity at one river point



Our river: the upper Godavari



Schematic · sewage figure: Govt. affidavit to NGT, 2023

Research & Key Insights

User & Context Research

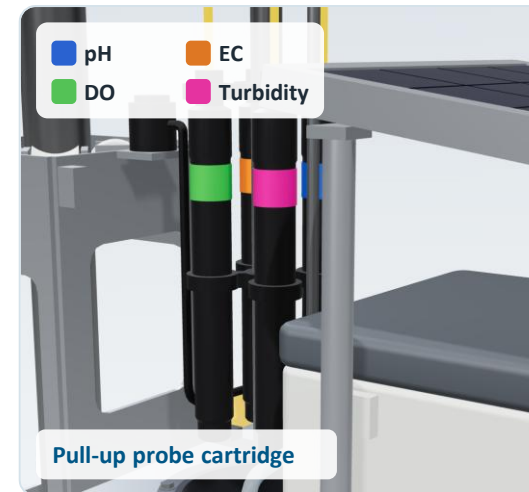
Water-works	time to close the intake
MPCB officer	a sealed sample for the lab
People at the ghat	a signal, without a phone
The river	2 m swings, weed, debris

Key Insights & Design Implications

Spills last hours	→ read every 15 min
Probes hint, labs prove	→ bottle a sample
Level swings 2 m	→ float on a guide pole
No phones at the ghat	→ a light anyone can read

Existing Solutions & Market Research

- 01 Hobby turbidity boards leak → **industrial RS485 probes**
- 02 Guide-pole pontoons work in marinas → **borrowed for the river**
- 03 Sensors + event detection flag illegal dumping



What the probes can see

MEASURE	CANNOT SEE
✓ pH	✗ Bacteria
✓ Conductivity	✗ Heavy metals
✓ Dissolved O ₂	✗ Pesticides
✓ Turbidity	
✓ Temperature	

So on an alert it bottles a sample for a certified lab

Design Requirements

Technical Requirements

15 min

reading interval

2 m

level travel, no motor

1.5 m/s + 500 N

flood current + debris hit

≥ 2×

reserve buoyancy

≥ 100 mm

freeboard

10+ days

on battery, no sun

Constraints

Budget

student-funded; off-the-shelf parts

Site

no riverbed work: clamps to a bridge pier

River

monsoon floods, silt, water hyacinth

Materials

rust-free: 316L, PE, anodised 6061

Human Factors

Service

from the deck, nobody wades in

Removal

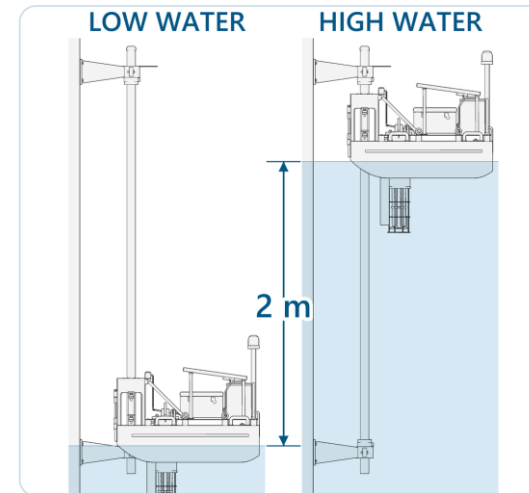
off the pole without tools

Probes

colour-coded, can't go in the wrong port

Handling

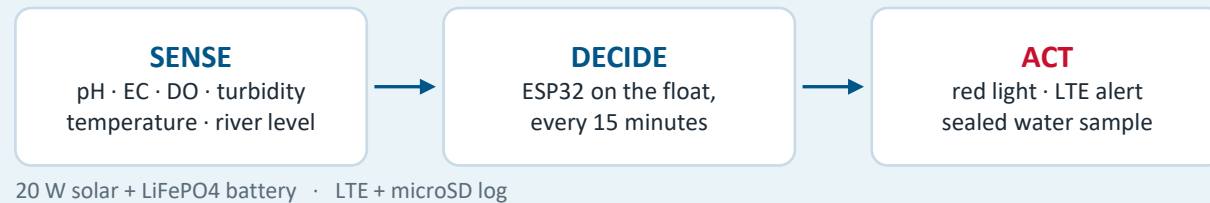
4 grab handles + 4 lifting eyes (48.5 kg)



Final Solution

Solution Overview

A solar float on a guide pole, clamped to a bridge pier. **The river moves it, not a motor.**

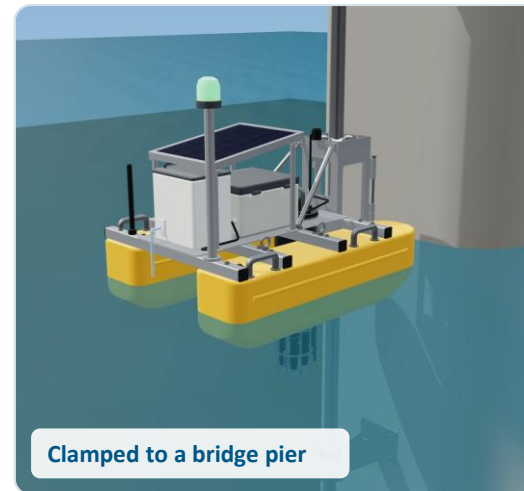
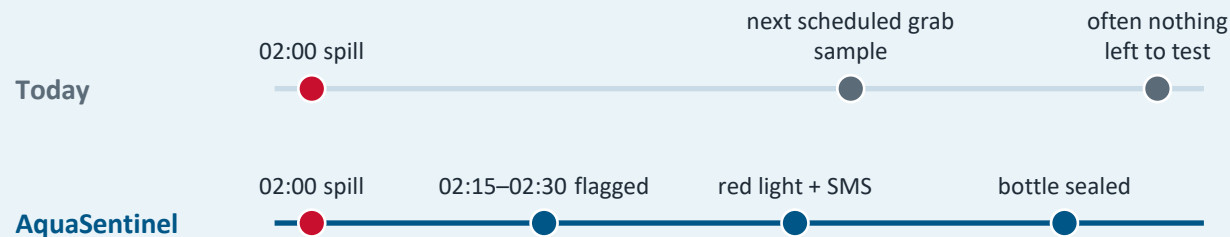


User Benefit

Water-works	A warning in time to close the intake
MPCB	A timed, sealed sample to act on
At the ghat	green = normal, red = stay out of the water

Problem-Solution Fit

Same spill, two outcomes



Innovation & Key Features

Key Features

- 01 Pole-guided float, no motor
- 02 Pull-up probe cartridge + wiper
- 03 Purge-then-fill sampler, 4 bottles
- 04 360° light: green / red
- 05 Tool-free carriage gate

Advantage Over Existing Solutions

every 15 min

vs a monthly grab sample

≈ ₹1.3 lakh*

vs ₹35 lakh+ per real-time station

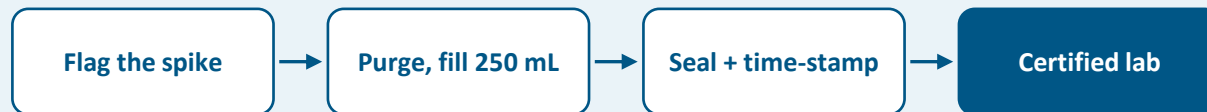
0 W

to hold station, vs a drone boat's battery

* prototype estimate

Innovation

It doesn't just measure. It keeps the evidence.



Final Product Design

AquaSentinel R3 at a glance

1.0 × 0.8 × 0.9 m

float size

48.5 kg

mass from CAD

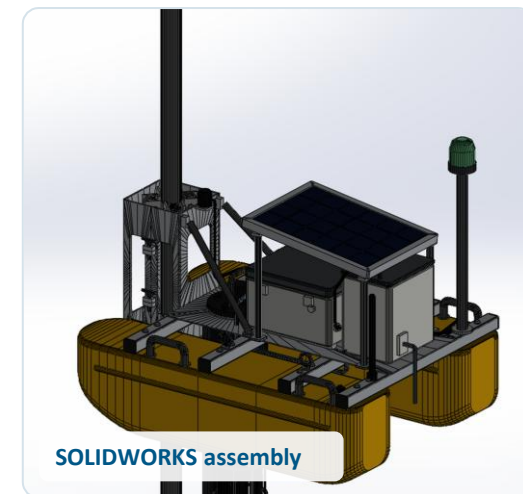
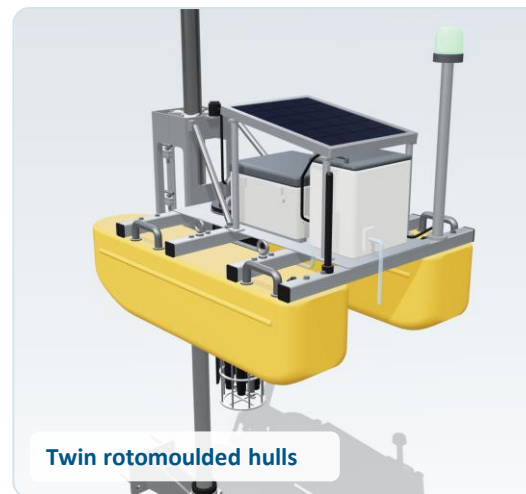
66 parts

native SOLIDWORKS

0 motors

the river moves it

Rotomoulded PE hulls · 6061 frame · 316L pole and brackets



Product Details & Functionality

Working Principle

Every 15 minutes, on the float itself. No cloud needed.



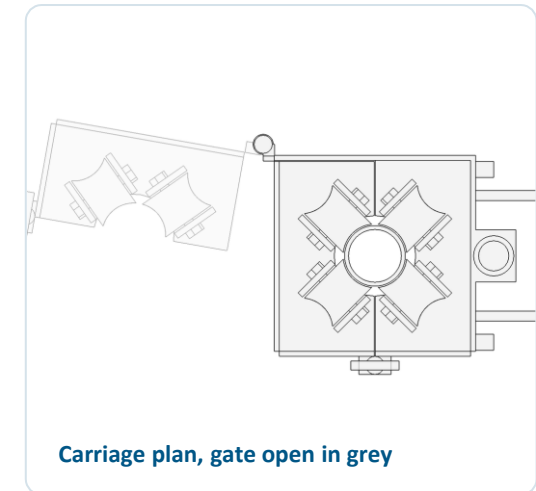
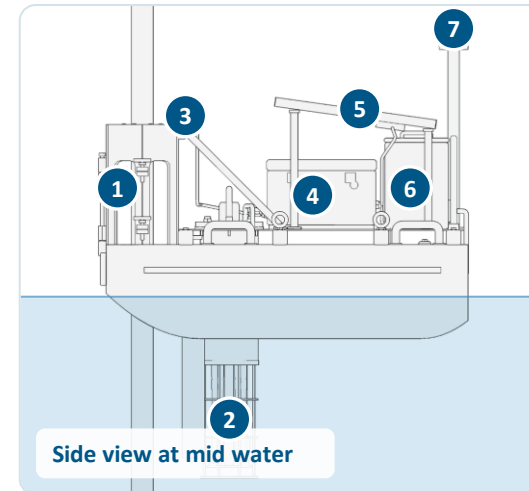
Hard limits act immediately · site baseline learned during first 7 days

Major Components

- 1 Guide pole + 8-roller carriage
- 2 Probes: pH · EC · DO · turbidity
- 3 Ultrasonic level sensor
- 4 Pod: ESP32 · LTE · microSD · battery
- 5 20 W solar panel
- 6 Sampler, 4 × 250 mL
- 7 Status light + LTE antenna

Mechanism / Product Structure

8 rollers	concave UHMW, 2 tiers 350 mm apart
1.5 mm	clearance on the 60.3 mm pole
Gate	swings open: hinge + 2 T-pins
Stops	rubber buffers end the 2 m travel



Design Validation

Validation Method

CALC Hand calcs on the CAD geometry: buoyancy, stability, jamming

FEA SOLIDWORKS Simulation: static, 316L, solid mesh

LOGIC Firmware logic replayed on 30 days of synthetic river data

Not tested in water yet. A tank test is next.

SOLIDWORKS Simulation

4.4

pole safety factor

38.3 MPa

peak stress, yield 170

1.3 mm

max pole sway

Bracket: 3.7 MPa peak, safety factor 46

Reactions equal the applied loads (871 N, 459 N)

Results

143 mm

freeboard

2.27×

reserve buoyancy

0.55 m

GM: rights itself to 60°

8.6×

anti-jam margin

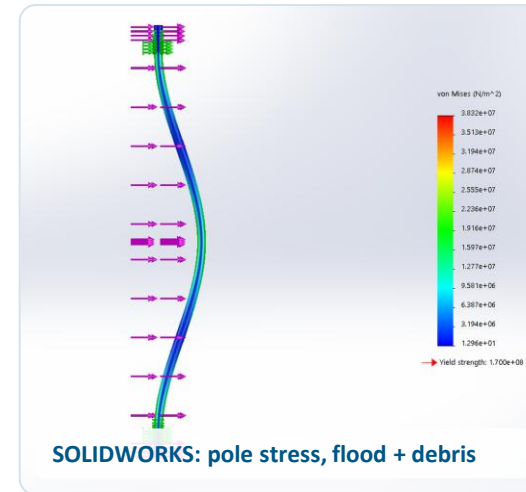
0

clashes over the 2 m travel

4 / 4

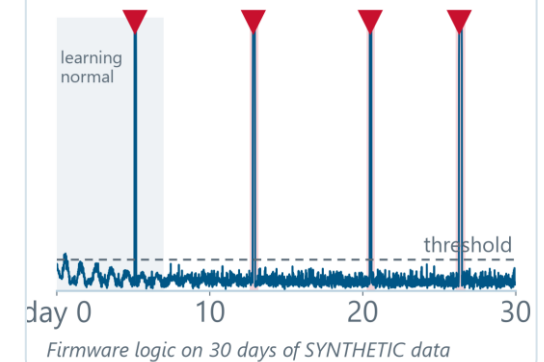
synthetic spill cases flagged*

** firmware logic replayed on synthetic river data, not field measurements*



4 of 4 spills flagged

how unusual each reading is (robust z)



Design Journey

Design Process



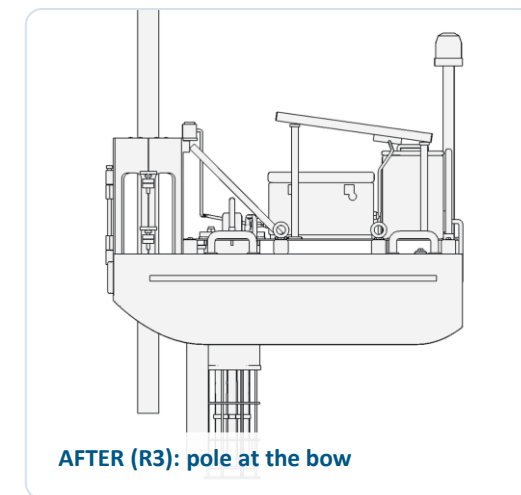
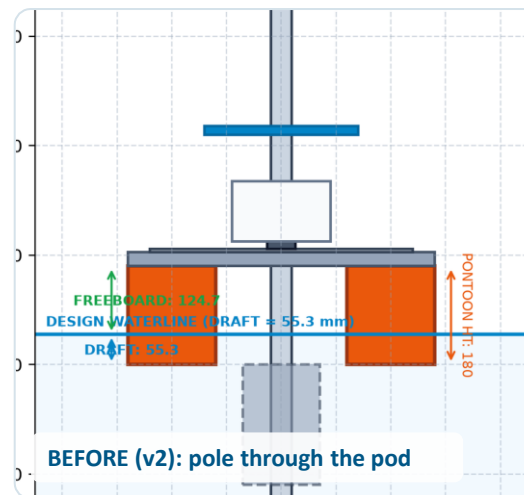
FEA in SOLIDWORKS Simulation. Each step broke something in the step before. That is how R3 got here.

What Changed from IDC?

Sliding collar	→ 8 rollers, ~5× less friction
1.5" tube	→ 2" Sch 40S pole
Hobby sensors	→ industrial RS485 probes
Depth sensor	→ level sensor looking up
No proof	→ + sampler, wiper, status light

Key Iterations

v2 → R3	Pole moved to the bow: in v2 it ran through the electronics box
25.4 → 48.5 kg	Real mass from CAD: hulls grew to 1000 × 230 × 260 mm
Free yaw	The float trails the current, so the pole takes no twist



Impact & Sustainability

Social Impact

96 readings a day

instead of one a month: earlier warning for people

who drink, bathe and farm with river water

A light anyone on the bank can read. No phone, no app.

Economic Impact

≈ 25 floats

for the price of one real-time monitoring station

₹1.34 lakh prototype estimate vs ₹35 lakh+ per station*

* real-time station cost benchmark: CSE

Environmental Impact

0 fuel · 0 motors

Solar only: nothing to leak into the river it watches

Short night-time dumps get recorded and sampled

Sustainability

Repair, don't replace

Probes and bottles swap out; one mould makes both hulls

316L, PE and aluminium are all recyclable

Viability & Future Development

Implementation & Manufacturing



Rotomoulded hulls (one mould) · welded 6061 carriage · off-the-shelf electronics

Current Limitations

- Not tested in water yet
- Probes give clues, not lab results
- 48.5 kg: a two-person lift
- Needs site flood data and permission

Cost & Market Potential

Prototype cost (estimate, quotes pending)

4 probes ₹81k

build ₹53k

= ₹1.34 lakh

Who buys

MPCB / CPCB

Water-works

Irrigation dept.

Industrial estates

Future Development

NEXT PHASE

Tank test: draft, 2 m travel, roller friction

PROPOSED

30-day Godavari pilot at Kopargaon (permit needed)

FUTURE SCALE-UP

A chain of floats along the river

Also next: SOLIDWORKS Flow Simulation to confirm drag

Final Project Summary

Problem

Spills pass in hours. Rivers are checked monthly.

People downstream find out too late, and there is nothing left to test.

Innovation

No motor, and it keeps the evidence.

The river lifts and turns it; on a spike it seals a time-stamped sample.

Solution

AquaSentinel: a solar float on a guide pole.

Reads the river every 15 min, decides on the float, then lights up, alerts and samples.

Impact

**Earlier warning + time-stamped evidence
≈ 1/25th the benchmark station cost.**

For water-works, people at the ghat and the pollution board.